

KIGOMA, Tanzania, United Republic ofWMO: **638010**Lat: **4.883S**Lon: **29.633E**Elev: **885**StdP: **91.14**Time Zone: **3.00 (E03)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	16.8	9.8	8.4	27.2	10.9	9.0	26.7	10.1	26.8	9.4	27.3	0.1	90	0.245

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	10.8	31.6	20.2	31.0	20.5	30.2	20.8	24.2	28.4	23.9	28.1	23.5	27.8	5.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	19.9	26.9	22.6	19.3	26.3	22.2	18.8	25.9	78.8	28.6	76.9	28.3	75.5	27.7	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.1	6.5	DB	14.1	33.5	1.2	0.8	13.2	34.0	12.5	34.5	11.8	34.9	11.0	35.5	(4)
(5)			WB	12.5	26.7	0.9	1.2	11.8	27.5	11.3	28.2	10.7	28.9	10.0	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	24.1	24.3	24.2	24.2	24.4	23.4	23.0	24.1	25.1	24.9	23.7	23.6	(6)	
	DBStd	1.17	0.88	1.04	1.12	1.04	0.91	0.89	0.93	0.95	1.02	1.22	1.22	0.89	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	5143	436	401	441	427	445	401	403	438	453	463	412	422	(10)	
	CDD18.3	2101	178	168	183	177	187	151	144	179	203	205	162	164	(11)	
	CDH23.3	14233	973	1016	1103	1077	1278	1143	1186	1531	1806	1540	824	756	(12)	
(13)	CDH26.7	3804	179	245	270	259	346	273	284	513	687	489	161	98	(13)	
(14)	Wind	WSAvg	1.8	1.4	1.5	1.5	1.6	1.8	2.0	2.1	2.2	2.3	1.9	1.5	1.3	(14)
(15) Precipitation	PrecAvg	978	129	117	126	167	54	5	2	5	17	66	148	141	(15)	
	PrecMax	1362	297	233	261	356	167	36	29	67	82	162	338	244	(16)	
	PrecMin	719	37	34	36	58	0	0	0	0	0	3	27	54	(17)	
	PrecStd	162	56	50	60	67	40	9	6	13	21	41	68	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	31.1	31.1	31.0	31.1	30.2	30.6	31.9	32.7	32.8	30.9	29.2	(19)	
		MCWB	23.2	23.4	23.3	22.4	21.4	19.7	18.7	18.8	19.5	20.1	21.4	23.1	(20)	
	2%	DB	29.1	29.9	30.0	29.9	30.1	29.8	29.4	30.8	31.5	31.2	29.2	28.5	(21)	
		MCWB	23.1	23.3	23.5	22.4	21.4	19.8	19.1	19.1	19.8	20.5	22.5	22.9	(22)	
	5%	DB	28.5	29.1	29.1	29.0	29.2	29.0	28.9	30.0	30.9	30.2	28.3	27.8	(23)	
		MCWB	22.9	23.2	23.1	22.7	21.7	20.3	19.4	19.4	20.0	21.0	22.5	22.7	(24)	
	10%	DB	27.9	28.2	28.3	28.2	28.7	28.2	28.2	29.2	30.1	29.4	27.4	27.0	(25)	
		MCWB	22.7	23.0	22.8	22.6	21.7	20.2	19.6	19.6	20.2	21.4	22.3	22.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.7	24.7	24.2	23.9	23.5	23.1	23.2	23.7	24.2	24.1	24.1	(27)	
		MCDB	28.2	28.7	29.2	28.4	28.1	27.7	27.8	27.9	28.6	28.7	28.1	28.1	(28)	
	2%	WB	23.8	24.1	24.1	23.6	23.1	22.5	22.1	22.4	22.9	23.3	23.3	23.5	(29)	
		MCDB	27.8	28.4	28.6	27.8	27.7	27.5	27.4	27.5	28.3	28.0	27.5	27.3	(30)	
	5%	WB	23.4	23.7	23.6	23.2	22.6	21.9	21.4	21.7	22.3	22.9	22.9	23.0	(31)	
		MCDB	27.3	28.0	28.0	27.5	27.2	27.0	26.6	27.1	27.8	27.5	27.0	26.7	(32)	
	10%	WB	23.0	23.2	23.1	22.8	22.2	21.2	20.6	21.0	21.8	22.4	22.5	22.6	(33)	
		MCDB	26.9	27.4	27.3	26.9	26.8	26.3	26.2	26.5	27.4	27.0	26.3	26.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.3	8.2	9.5	11.1	12.0	11.7	10.8	8.9	7.3	7.1	(35)	
		MCDBR	8.7	9.3	9.5	9.7	10.5	11.7	12.5	12.6	12.1	10.8	8.9	8.4	(36)	
	5% WB	MCWBR	4.1	4.3	4.3	3.9	3.8	4.5	5.1	5.2	4.9	4.2	4.0	3.9	(37)	
		MCWBR	8.1	8.7	9.1	8.7	9.4	10.3	11.1	10.8	10.0	8.7	8.1	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.463	0.438	0.403	0.361	0.380	0.407	0.450	0.488	0.470	0.448	0.405	0.428	(40)	
		taud	2.258	2.349	2.472	2.605	2.488	2.357	2.203	2.103	2.185	2.255	2.410	2.359	(41)	
	Ebn at Noon		878	900	920	935	890	849	816	808	848	880	924	906	(42)	
		Edh at Noon	145	133	116	98	105	117	139	160	152	144	123	130	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.84	5.21	5.20	5.12	5.22	5.38	5.67	5.80	5.87	5.35	4.53	4.56	(44)	
	RadStd		0.27	0.29	0.32	0.40	0.34	0.34	0.21	0.32	0.35	0.33	0.37	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page